

Feature Selection in Statistical Machine Learning

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Lecture 17

Recap and outline

- What we've covered so far and plan for remainder of course
- Interpretable/explainable machine learning
- Motivation and outline for interpretable machine learning

What we've covered so far

- Module 1: Feature selection definitions
- Module 2: Feature selection methods
- Module 3: Feature selection and graphical models
- Module 4 (to come): Feature selection and interpretable/explainable machine learning

Feature selection definitions

Data: (p features and n samples)

$$(\{X_j^{(i)}\}_{j=1}^p, Y^{(i)})_{i=1}^n$$

Model/function:

$$\mathbb{E}[Y|X_1, X_2, \dots, X_p] = f(X_1, X_2, \dots, X_p)$$

Feature selection definitions:

- Statistical risk-based definition:

$$S^* \in \arg \min_{S \subseteq \{1, 2, \dots, p\}, |S| \leq s} \min_{f \in \mathcal{F}} \mathbb{E}[\mathcal{L}(Y, f(X_S))]$$

- Conditional independence definition:

$$S^* \in \arg \min_{S \subseteq \{1, 2, \dots, p\}, Y \perp X_{S^c} | X_S} |S|$$

- Local feature-wise definition:

$$Y \perp X_j | X_{-j}, \text{ for all } j \in \tilde{S}^c$$

$$Y \not\perp X_j | X_{-j}, \text{ for all } j \in \tilde{S}$$

Feature selection methods

- Filter methods (e.g. correlation, mutual information, conditional correlation screening)
- Embedded methods (e.g. OLS+p-values, Lasso, MDI)
- Wrapper methods (LOCO, AIC, BIC, forward and backward selection)
- Validation methods (prediction, stability, e.t.c.)

What are the biggest challenges in feature selection?

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- Dependent features
- Non-linearity
- Computation

Feature selection and graphical models

- Undirected graphical models
 - ▶ Natural for spatial (non-directional) dependence
 - ▶ Neighborhood selection for each of the p features equivalent to feature selection
 - ▶ Need to resolve inconsistent pair-wise neighborhoods
- Directed acyclic graphical (DAG) models
 - ▶ Natural for temporal or causal structural equation model (directional) dependence
 - ▶ All DAGs have a (possibly non-unique) causal ordering
 - ▶ Parent selection for each of the p features equivalent to feature selection

Plan for the remainder of the semester

- 8 lectures for remainder of course
- Next 5-6 lectures, final module on interpretable/explainable machine learning
- 7th lecture exam preparation
- April 29 (8th lecture) final exam
- Project due at the end of semester (April 29)
- Homework 3 due Thursday April 9
- Homework 4 due Thursday April 30

Module: Feature selection and interpretable/explainable machine learning

- What is it and why do we care?
- Limitations of feature selection in modern machine learning
 - Interactions
 - Local (individual samples) vs global (averaged across samples)
- Intrinsic interpretability vs post-hoc interpretability and connections to feature selection
- Feature attribution (this week)
- Shapley value framework (next week)
- Shapley values and feature selection (week after)

What is interpretability/explainability?

What is interpretability/explainability?

Fundamental humanistic concept making it more possible for humans to understand why predictions are made

Addresses question like:

- Why models make certain predictions?
- Which features are the most important?
- If you change this feature/attribute, how would the overall prediction change?

Note: Very contextual and human-driven

Interpretability setup (similar to feature selection)

Data: (p features and n samples)

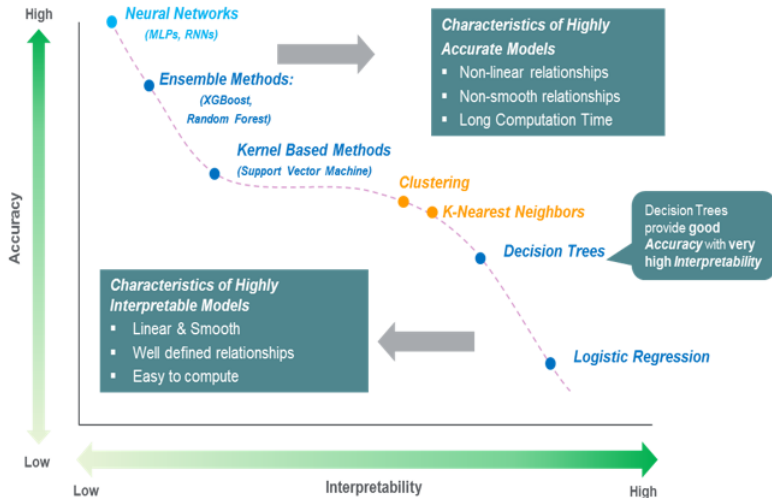
$$(\{X_j^{(i)}\}_{j=1}^p, Y^{(i)})_{i=1}^n$$

Model/function:

$$\mathbb{E}[Y|X_1, X_2, \dots, X_p] = f(X_1, X_2, \dots, X_p)$$

- f may be linear function, tree/forest-based model, neural network, e.t.c.
- f maybe black-box foundation model like large language models

Interpretability vs accuracy chart



Interpretability and feature selection

- Until recently (maybe the past 10-20 years), interpretable machine learning was more or less equivalent to feature selection
- Classically smaller datasets, mainly tabular and main goal was scientific discovery
- More recently large-scale black-box and foundation models, feature selection does not directly correspond to interpretability
- Local/individualized explanations become critical

Why do we care about interpretability?

Why do we care about interpretability?

- Humans ideally need to understand why outputs are produced
- Builds trust in the systems/predictions
- Bias and fairness
- Transparency

Example 1: Sentiment prediction in language

Goal: Determine whether the sentence is positive, negative or neutral.

"The movie was good" - positive, easy

"The movie was not good" - positive, wrong

- The word "not" is very confusing (negation issue)
- Interpretability reveals contribution of each word to sentiment prediction, "good" very high, "movie" neutral, and "not" very small
- Interpretability explains why this model fails (transparency)
- Negation failure can be much more critical in higher-stakes settings (e.g. medical diagnosis)
- Modern LLMs are significantly better at overcoming this failure, but first the failure needed to be detected through interpretability tools

Example 2: Medical diagnosis, risk of heart attack

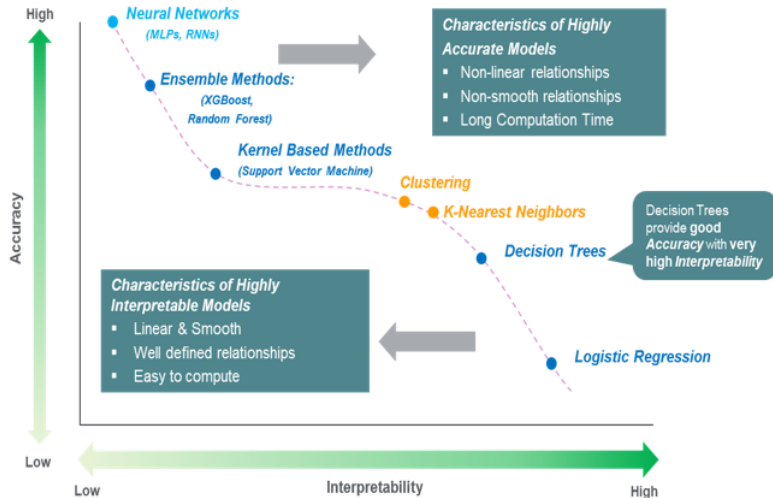
Goal: Predict heart attack risk for individual patients based on the following features: age, cholesterol, blood pressure, smoking status, family history.

- Feature selection/ranking reveals that age is most important, cholesterol is moderately important and smoking status is not important
- Consider specific example: 45-year old patient with normal cholesterol (180), slightly high blood pressure and is a smoker
- When we look at local attribution (to be discussed shortly), age slightly higher risk, cholesterol not important, smoking is hugely significant, blood pressure moderate
- **Reason:** interaction between blood pressure and smoking status significant
- Highlights two issues with feature selection: local vs global and interaction effects

Intrinsic interpretability vs post-hoc interpretability

- Often distinctions made between two settings: intrinsically interpretable models/algorithms and post-hoc explanations
- "Simpler" models like linear/logistic regression and decision trees fall into first category while more complex models/algorithms like random forests, neural networks, LLMs, e.t.c. require the second approach
- Will often refer to this distinction going forward
- Also a nice (slightly vague connection) to concepts we've learned so far in feature selection

Recall: Chart



Intrinsic interpretability and embedded feature selection methods

- Models that are designed to be understandable like linear/logistic regression, decision trees, other rule-based systems
- Intrinsic interpretability arises from the fact that models/quantities used for the prediction process have a natural interpretation (e.g. co-efficients, MDI)
- Recall that embedded methods in feature selection relate to determining if these quantities are statistically significant (e.g. Lasso, MDI, e.t.c.)

Post-hoc interpretability and wrapper feature selection methods

- Used for complex black-box models like neural networks, random forests to an extent, e.t.c.
- For these models/algorithms prediction does not directly contain interpretable quantities
- Need to use post-hoc methods that treat methods like a black-box (this will be one of the main focuses of this module)
- Recall wrapper methods for feature selection (e.g. LOCO, forward/backward selection, e.t.c.) can be viewed as a post-hoc interpretability method

Feature analysis: Global vs local

- One of the key conceptual shifts is from global inference/statements (what is significant across this dataset), what is happening for a specific instance/set of instances
- Local has been used in lots of different contexts. Here we specifically mean local with respect to *samples*
- Medical diagnosis is standard setting where this matters (diagnosing specific patient rather than averaging across a dataset)
- Feature selection/ranking reveals that age is most important, cholesterol is moderately important and smoking status is not important
- Consider specific example: 45-year old patient with normal cholesterol (180), slightly high blood pressure and is a smoker

Feature selection (in this course) is implicitly global

Recall feature selection definition:

- Statistical risk-based definition:

$$S^* \in \arg \min_{S \subseteq \{1,2,\dots,p\}, |S| \leq s} \min_{f \in \mathcal{F}} \mathbb{E}[\mathcal{L}(Y, f(X_S))]$$

- Conditional independence definition:

$$S^* \in \arg \min_{S \subseteq \{1,2,\dots,p\}, Y \perp X_{S^c} | X_S} |S|$$

- Implicitly assumes a distribution over the dataset
- Does not say anything about impacts for individual (local) predictions
- We shift our focus for now to local explanations rather than global importance

How do we get local explanations? Feature attribution

Data: (p features and n samples)

$$(\{X_j^{(i)}\}_{j=1}^p, Y^{(i)})_{i=1}^n$$

Model/function:

$$\mathbb{E}[Y|X_1, X_2, \dots, X_p] = f(X_1, X_2, \dots, X_p)$$

Prediction algorithm/estimator:

$$\widehat{f}(\cdot) = \mathcal{A}((\{X_j^{(i)}\}_{j=1}^p, Y^{(i)})_{i=1}^n)$$

- For individual with features $\tilde{X} = (\tilde{X}_1, \tilde{X}_2, \dots, \tilde{X}_p)$
- Overall prediction is $\widehat{f}(\tilde{X}_1, \tilde{X}_2, \dots, \tilde{X}_p)$
- How do we assign an attribution ϕ_j for each feature $\tilde{X}_j$ that captures importance for the *individual prediction*?
- Key property (completeness/efficiency):

$$\widehat{f}(\tilde{X}_1, \tilde{X}_2, \dots, \tilde{X}_p) = \sum_{j=1}^p \phi_j$$

- Prediction can be decomposed into sum of p feature attributions

Feature attribution example: Linear regression

Data: (p features and n samples)

$$(\{X_j^{(i)}\}_{j=1}^p, Y^{(i)})_{i=1}^n$$

Model/function:

$$\mathbb{E}[Y|X_1, X_2, \dots, X_p] = \sum_{j=1}^p \beta_j X_j$$

Prediction algorithm/estimator:

$$\widehat{f}(\cdot) = \mathcal{A}(\{X_j^{(i)}\}_{j=1}^p, Y^{(i)})_{i=1}^n = (\widehat{\beta}_1, \widehat{\beta}_2, \dots, \widehat{\beta}_p)$$

- For individual with features $\tilde{X} = (\tilde{X}_1, \tilde{X}_2, \dots, \tilde{X}_p)$
- Overall prediction is $\widehat{f}(\tilde{X}_1, \tilde{X}_2, \dots, \tilde{X}_p) = \sum_{j=1}^p \widehat{\beta}_j \tilde{X}_j$
- How do we assign an attribution ϕ_j for each feature $\tilde{X}_j$ that captures importance for the *individual prediction*?

Feature attribution example: Linear regression cont.

- For individual with features $\tilde{X} = (\tilde{X}_1, \tilde{X}_2, \dots, \tilde{X}_p)$
- Overall prediction is $\widehat{f}(\tilde{X}_1, \tilde{X}_2, \dots, \tilde{X}_p) = \sum_{j=1}^p \widehat{\beta}_j \tilde{X}_j$
- How do we assign an attribution ϕ_j for each feature $\tilde{X}_j$ that captures importance for the *individual prediction*?
- **Feature attribution:** $\phi_j = \widehat{\beta}_j \tilde{X}_j$
- Key property (completeness/efficiency):

$$\widehat{f}(\tilde{X}_1, \tilde{X}_2, \dots, \tilde{X}_p) = \sum_{j=1}^p \phi_j$$

- Satisfied for $\phi_j = \widehat{\beta}_j \tilde{X}_j$